

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 5x + 1}{x + 2} \quad \boxed{6}$$

$$\lim_{x \rightarrow +\infty} \frac{\sqrt{x^2 + 1}}{x} \quad \boxed{7}$$

تمرين 3

احسب النهايات التالية :

$$\lim_{x \rightarrow 4} \frac{x^2 - 8x + 16}{x^2 - 5x + 4} \quad \boxed{1}$$

$$\lim_{x \rightarrow 3} \frac{x - 3}{x^2 - 6x + 9} \quad \boxed{2}$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x^2 - x - 6} \quad \boxed{3}$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1} \quad \boxed{4}$$

$$\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1} \quad \boxed{5}$$

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2 + x - 1}{x^2 - 1} \quad \boxed{6}$$

$$\lim_{x \rightarrow 1} \frac{x^3 - 3x + 2}{x^4 - 4x + 3} \quad \boxed{7}$$

$$\lim_{x \rightarrow 2} \frac{x - 2}{x^2 + x - 6} \quad \boxed{8}$$

$$\lim_{x \rightarrow -1} \frac{x^3 + 1}{x^2 + x} \quad \boxed{9}$$

$$\lim_{x \rightarrow 2} \frac{x^4 - 16}{x^3 - 8} \quad \boxed{10}$$

$$\lim_{x \rightarrow 5} \frac{\sqrt{x} - \sqrt{5}}{x - 5} \quad \boxed{11}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{4x} - 2}{x^2 - 1} \quad \boxed{12}$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1-x} - 1}{\sqrt{1+x} - 1} \quad \boxed{13}$$

$$\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - 2}{x - 2} \quad \boxed{14}$$

$$\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - 3}{\sqrt{3x} - 3} \quad \boxed{15}$$

تمرين 1

احسب النهايات التالية :

$$\lim_{x \rightarrow 3} \frac{4x - 1}{x - 3} \quad \boxed{1}$$

$$\lim_{x \rightarrow 2} \frac{-x + 1}{x^2 - 5x + 6} \quad \boxed{2}$$

$$\lim_{x \rightarrow 3} \frac{x}{x - 3} \quad \boxed{3}$$

$$\lim_{x \rightarrow 4} \frac{x}{x - 4} \quad \boxed{4}$$

$$\lim_{x \rightarrow 4} \frac{\sqrt{2} + 2}{4 - x} \quad \boxed{5}$$

$$\lim_{x \rightarrow 1} \frac{2x - 1}{x - 1} \quad \boxed{6}$$

$$\lim_{x \rightarrow 1} \frac{x^2 - 4}{|x - 1|} \quad \boxed{7}$$

$$\lim_{x \rightarrow 2} \frac{-x + 5}{|x - 2|} \quad \boxed{8}$$

$$\lim_{x \rightarrow 2} \frac{x^2 + 1}{x^2 - 3x + 2} \quad \boxed{9}$$

$$\lim_{x \rightarrow 1} \frac{x^2 + x + 5}{\sqrt{x} - 1} \quad \boxed{10}$$

$$\lim_{x \rightarrow 5} \frac{-2x + 1}{(x - 5)^2} \quad \boxed{11}$$

تمرين 2

احسب النهايات التالية :

$$\lim_{x \rightarrow +\infty} \frac{1 - x^2 + 3x}{1 + x - x^2} \quad \boxed{1}$$

$$\lim_{x \rightarrow +\infty} \frac{5x^3 - 3x^2 + 7}{2x^4 + 3x^2 + 1} \quad \boxed{2}$$

$$\lim_{x \rightarrow +\infty} \frac{-3x}{\sqrt{x^2 - 2x}} \quad \boxed{3}$$

$$\lim_{x \rightarrow +\infty} x^2 - \sqrt{x^4 + 2x^2 - 1} \quad \boxed{4}$$

$$\lim_{x \rightarrow +\infty} \frac{2x^2 - 1}{x^2 - \sqrt{x^4 + 2x^2 - 1}} \quad \boxed{5}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + 2x - 1} - \sqrt{2x^2 - x + 1} \quad \mathbf{6}$$

$$\lim_{x \rightarrow +\infty} \sqrt{3x^2 + 2x + 2} - \sqrt{3x + 1} \quad \mathbf{7}$$

$$\lim_{x \rightarrow +\infty} \frac{3x - \sqrt{x^2 + x + 1}}{x + 3} \quad \mathbf{8}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x + 1} - x \quad \mathbf{9}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + x + 3} - 4x \quad \mathbf{10}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + x - 2} - 2x \quad \mathbf{11}$$

$$\lim_{x \rightarrow 0} \frac{1}{x} - \frac{1}{\sqrt{x}} \quad \mathbf{12}$$

تمرين 6

احسب النهايات التالية :

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x} \quad \mathbf{1}$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{x} \quad \mathbf{2}$$

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{3x} \quad \mathbf{3}$$

$$\lim_{x \rightarrow 0} \frac{3x}{\sin 2x} \quad \mathbf{4}$$

$$\lim_{x \rightarrow 0} \frac{4x}{\sin 3x} \quad \mathbf{5}$$

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{4x} \quad \mathbf{6}$$

$$\lim_{x \rightarrow 0} \frac{\tan 5x}{4x} \quad \mathbf{7}$$

$$\lim_{x \rightarrow 0} \frac{5x}{\tan 4x} \quad \mathbf{8}$$

$$\lim_{x \rightarrow 0} \frac{2x}{\tan 3x} \quad \mathbf{9}$$

تمرين 7

احسب النهايات التالية :

$$\lim_{x \rightarrow +\infty} \frac{\sin x}{1 + x^2} \quad \mathbf{1}$$

$$\lim_{x \rightarrow -\infty} \frac{1 + x + \cos x}{x - 1} \quad \mathbf{2}$$

$$\lim_{x \rightarrow +\infty} \frac{x^2 + 1 \sin x}{x} \quad \mathbf{3}$$

$$\lim_{x \rightarrow 1} \frac{2x - \sqrt{x + 3}}{\sqrt{x^2 + 3} - 2} \quad \mathbf{16}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{x - \frac{\pi}{2}} \quad \mathbf{17}$$

$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{2 \cos x - 1}{x - \frac{\pi}{3}} \quad \mathbf{18}$$

$$\lim_{x \rightarrow a} \frac{\sqrt{x} - \sqrt{a} - \sqrt{x - a}}{\sqrt{x^2 - a^2}} \quad (a > 0) \quad \mathbf{19}$$

تمرين 4

احسب النهايات التالية :

$$\lim_{x \rightarrow 0} x \left(\sqrt{4 + \frac{1}{x}} - 2 \right) \quad \mathbf{1}$$

$$\lim_{x \rightarrow -\infty} \frac{4}{x} \left(\frac{1}{x^2} + \frac{1}{x} \right) (x^2 + 1) \quad \mathbf{2}$$

$$\lim_{x \rightarrow +\infty} \left(\frac{1}{x} - \frac{1}{\sqrt{x}} \right) (\sqrt{x} - 1) \quad \mathbf{3}$$

$$\lim_{x \rightarrow +\infty} \frac{-2}{x} (2\sqrt{x} - 1) \quad \mathbf{4}$$

$$\lim_{x \rightarrow +\infty} \frac{4}{x} (1 - \sqrt{x}) \quad \mathbf{5}$$

$$\lim_{x \rightarrow +\infty} \frac{1}{x^2} - \frac{1}{x} (x^2 + x) \quad \mathbf{6}$$

$$\lim_{x \rightarrow -\infty} \frac{1}{x^2} + \frac{1}{x} (x^3 + 1) \quad \mathbf{7}$$

$$\lim_{x \rightarrow 0} x \sqrt{\frac{1 - x}{x}} \quad \mathbf{8}$$

تمرين 5

احسب النهايات التالية :

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + 1} - x \quad \mathbf{1}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + 1} - 3x \quad \mathbf{2}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + x - 2} - 2x \quad \mathbf{3}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 + 2x - 1} - x \quad \mathbf{4}$$

$$\lim_{x \rightarrow +\infty} \sqrt{x^2 - 1} - x \quad \mathbf{5}$$